

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Numerical

$\lim_{x \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to

Q2: 24 Feb (Shift 2) - Single Correct

A possible value of $\tan \left(\frac{1}{4} \sin^{-1} \frac{\sqrt{63}}{8} \right)$ is:

(1) $\frac{1}{2\sqrt{2}}$

(2) $\frac{1}{\sqrt{7}}$

(3) $\sqrt{7} - 1$

(4) $2\sqrt{2} - 1$

Q3: 25 Feb (Shift 2) - Single Correct

$\operatorname{cosec} \left[2 \cot^{-1}(5) + \cos^{-1} \left(\frac{4}{5} \right) \right]$ is equal to:

(1) $\frac{75}{56}$

(2) $\frac{65}{56}$

(3) $\frac{56}{33}$

(4) $\frac{65}{33}$

Q4: 26 Feb (Shift 1) - Single Correct

If $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}$; $0 < x < 1$, then the value of $\cos \left(\frac{\pi c}{a+b} \right)$ is:

(1) $\frac{1-y^2}{2y}$

(2) $\frac{1-y^2}{1+y^2}$

(3) $1 - y^2$

(4) $\frac{1-y^2}{y\sqrt{y}}$

Q5: 26 Feb (Shift 2) - Single Correct

Questions with Answer Keys

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If $0 < a, b < 1$, and $\tan^{-1} a + \tan^{-1} b = \frac{\pi}{4}$, then the value of $(a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \dots$

is :

(1) $\log_e 2$

(2) $\log_e \left(\frac{e}{2}\right)$

(3) e

(4) $e^2 - 1$

Answer Key**Q1 (1)****Q2 (2)****Q3 (2)****Q4 (2)****Q5 (1)**

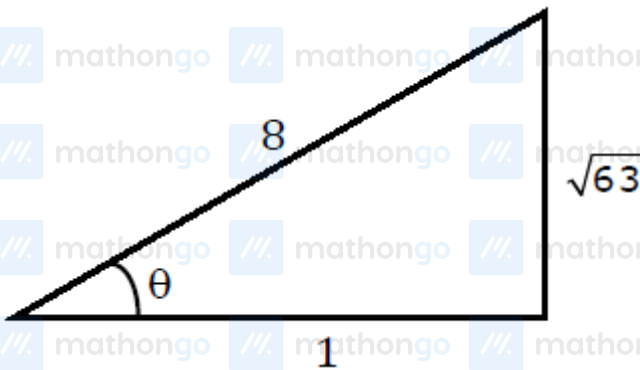
Q1 (1)

$$\begin{aligned} & \tan\left(\lim_{n \rightarrow \infty} \sum_{r=1}^n [\tan^{-1}(r+1) - \tan^{-1}(r)]\right) \\ &= \tan\left(\lim_{n \rightarrow \infty} \left(\tan^{-1}(n+1) - \frac{\pi}{4}\right)\right) \\ &= \tan\left(\frac{\pi}{4}\right) = 1 \end{aligned}$$

Q2 (2)

$$\tan\left(\frac{1}{4} \sin^{-1} \frac{\sqrt{63}}{8}\right)$$

$$\text{Let } \sin^{-1}\left(\frac{\sqrt{63}}{8}\right) = \theta \quad \sin \theta = \frac{\sqrt{63}}{8}$$



$$\cos \theta = \frac{1}{8}$$

$$2 \cos^2 \frac{\theta}{2} - 1 = \frac{1}{8}$$

$$\cos^2 \frac{\theta}{2} = \frac{9}{16}$$

$$\cos \frac{\theta}{2} = \frac{3}{4}$$

$$\frac{1 - \tan^2 \frac{\theta}{4}}{1 + \tan^2 \frac{\theta}{4}} = \frac{3}{4}$$

$$\tan \frac{\theta}{4} = \frac{1}{\sqrt{7}}$$

Q3 (2)

$$\operatorname{cosec}\left(2 \cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right)$$

$$\operatorname{cosec}\left(2 \tan^{-1}\left(\frac{1}{5}\right) + \cos^{-1}\left(\frac{4}{5}\right)\right)$$

$$= \operatorname{cosec}\left(\tan^{-1}\left(\frac{2\left(\frac{1}{5}\right)}{1 - \left(\frac{1}{5}\right)^2}\right) + \cos^{-1}\left(\frac{4}{5}\right)\right)$$

$$= \operatorname{cosec}\left(\tan^{-1}\left(\frac{5}{12}\right) + \cos^{-1}\left(\frac{4}{5}\right)\right)$$

Hints and Solutions

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$$\text{Let } \tan^{-1}(5/12) = \theta \Rightarrow \sin \theta = \frac{5}{13}, \cos \theta = \frac{12}{13}$$

$$\text{and } \cos^{-1}\left(\frac{4}{5}\right) = \phi \Rightarrow \cos \phi = \frac{4}{5} \text{ and } \sin \phi = \frac{3}{5}$$

$$= \operatorname{cosec}(\theta + \phi)$$

$$= \frac{1}{\sin \theta \cos \phi + \cos \theta \sin \phi}$$

$$= \frac{1}{\frac{5}{13} \cdot \frac{4}{5} + \frac{12}{13} \cdot \frac{3}{5}} = \frac{65}{56}$$

Q4 (2)

$$\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}$$

$$\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\sin^{-1} x + \cos^{-1} x}{a+b} = \frac{\pi}{2(a+b)}$$

$$\text{Now, } \frac{\tan^{-1} y}{c} = \frac{\pi}{2(a+b)}$$

$$2 \tan^{-1} y = \frac{\pi c}{a+b}$$

$$\Rightarrow \cos\left(\frac{\pi c}{a+b}\right) = \cos(2 \tan^{-1} y) = \frac{1-y^2}{1+y^2}$$

Q5 (1)

$$\tan^{-1}\left(\frac{a+b}{1-ab}\right) = \frac{\pi}{4} \Rightarrow a+b = 1-ab \Rightarrow (1+a)(1+b) = 2$$

$$\text{Now, } (a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) \dots \infty$$

$$= \left(a - \frac{a^2}{2} + \frac{a^3}{3} \dots\right) + \left(b - \frac{b^2}{2} + \frac{b^3}{3} \dots\right)$$

$$\log_e(1+a) + \log_e(1+b) = \log_e(1+a)(1+b) = \log_e 2$$